

Navid Rezazadeh

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Summary

Machine learning research engineer with a Ph.D. in optimization, stochastic systems, and control. Research spans time-series signal modeling, generative distribution modeling, verification, distributed optimization, and high-performance scientific Python/CUDA for quantitative research.

Technical Skills

ML & Statistics: Transformers, time-series forecasting, walk-forward validation, ARIMA/ARMA, diffusion models, GAN/VAE, copulas, Kalman/sigma-point/particle filtering, MMD, PCA/SVD, regression

Mathematics: Stochastic processes, optimization, submodular optimization, graph algorithms, control theory, estimation, linear algebra, numerical simulation

Tools & Frameworks: Python, PyTorch, CUDA, multi-GPU orchestration, TensorFlow, NumPy, pandas, CuPy, batched linear algebra, MATLAB, C/C++, Git

Experience

Apple

Machine Learning Research Engineer

San Diego, CA

Jun 2022 – Present

- Vectorized scalar scientific pipelines into batched PyTorch/CUDA with multi-GPU data parallelism, reducing parameter sweeps from multi-week scale to sub-hour while preserving CPU-reference parity.
- Research workstreams: Foundation Models for Sensor and Time-Series Intelligence; Generative Simulation and Distributional Certification; Neural Inference and Calibration; High-Performance Scientific ML Infrastructure.

University of California, Irvine

Graduate Research Assistant

Irvine, CA

Jun 2016 – Jun 2022

- Developed optimization and control algorithms for distributed decision making, privacy-preserving estimation, and submodular maximization with provable guarantees.
- Integrated neural networks with contraction theory for stable dynamics and control policies; built Kalman, sigma-point, and particle filtering methods for probabilistic localization and state estimation.

Selected Projects

Signal Forecasting Under Stochastic Drift

Transformers, Informer, ARIMA/ARMA

- Short-horizon forecasting with rolling baselines, temporal features, and outlier-aware evaluation; compared Transformer/Informer models against ARIMA/ARMA statistical baselines under rolling out-of-sample validation.

Generative Modeling for Scientific Simulators

Diffusion, GANs, Gaussian Copulas, MMD

- Learned conditional samplers and simulator surrogates; validated generated samples with feature statistics, residual tests, chi-square checks, MMD, and error-driven candidate selection.

GPU-Accelerated Scientific ML Pipeline

PyTorch, CUDA, Multi-GPU, Batched Linalg

- Re-architected scalar sensor-to-physical inference with batched weighted least squares, pseudoinverse, broadcasted geometric transforms, cached GPU state, adaptive batching, and multi-GPU execution; reduced per-run evaluation from hours to seconds.

Selected Publications

- [1] *IEEE TCNS, 2024*
N. Rezazadeh, S.S. Kia, “A Study of Privacy Preservation in Average Consensus Algorithm via Deterministic Obfuscation Signals.”
- [2] *Automatica, 2023*
N. Rezazadeh, S.S. Kia, “Distributed Strategy Selection: A Submodular Set Function Maximization Approach.”
- [3] *IEEE RA-L, 2022 – NeurIPS SafeRL Best Paper Award*
N. Rezazadeh, M. Kolarich, S.S. Kia, N. Mehr, “Learning Contraction Policies From Offline Data.”
- [4] *Automatica, 2021*
N. Rezazadeh, S.S. Kia, “A Sub-Modular Receding Horizon Solution for Mobile Multi-Agent Persistent Monitoring.”
- [5] *IEEE CDC, 2022*
N. Rezazadeh, S.S. Kia, “Distributed Submodular Maximization: Trading Performance for Privacy.”
- [6] *IEEE CDC, 2021*
N. Rezazadeh, S.S. Kia, “Multi-Agent Maximization of a Monotone Submodular Function via Maximum Consensus.”

Education

University of California, Irvine <i>Ph.D. Mechanical & Aerospace Engineering, GPA 3.99/4.00</i> • Thesis: Distributed Strategy Selection Over Graphs: Optimality and Privacy	2017 – 2022
University of California, Irvine <i>M.S. Mechanical & Aerospace Engineering, GPA 4.00/4.00</i>	2016 – 2017
Sharif University of Technology <i>B.S. Mechanical Engineering, GPA 3.85/4.00</i> <i>Selected coursework:</i> Algorithms; Stochastic Processes; Real Analysis; Bayesian Data Analysis; Deep Learning and Sequence Models; Advanced Estimation and Detection; Classification, Parameter Estimation, and Filtering.	2010 – 2014

Honors

NeurIPS SafeRL Best Paper Award Holmes Fellowship, UCI MAE Samueli Endowed Fellowship, UCI Iran National Elite Foundation Fellowship Ranked 250th among 300,000+ in Iran national university entrance exam
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